

Clinical and Physiological Data from PH Monitoring Study

Background

Fit-PH is a dataset derived from *PHoenix*, a Phase IV randomized clinical trial involving patients with pulmonary hypertension (PH). The trial's primary aim was to assess dose-response effects and clinical efficacy of two medications: **riociguat** and **selexipag**, in addition to standard care. As part of the study, participants were also implanted with cardiovascular monitoring devices to enable continuous remote capture of physiological signals.

The dataset was collected with the aim of developing personalized treatment-response profiles, by correlating each individual's device-measured physiology with medication dosing and outcomes. Remote monitoring detects early signs of therapeutic response -or adverse changes- to enable timely adjustment strategies.

The patient cohort comprises of 17 individuals diagnosed with PH who enrolled in the clinical trial and were randomised to receive varying doses of riociguat or selexipag. Patients were monitored over a 29 week period and were required to attend 4 hospital visits. During each visit, they underwent clinical assessments for physiological, functional and psychological wellbeing.

Additional data were also recorded remotely from home, through self-administered tests and implantable devices that tracked continuous physiological signals e.g. blood pressure. Clinical trial outcomes - functional status, adverse events were linked to both in-clinic and remotely acquired measurements.

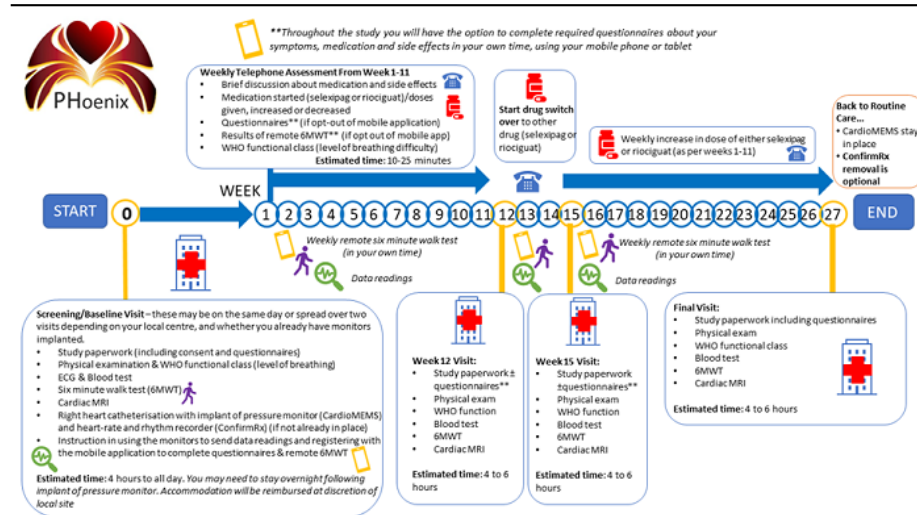


Figure 1: Summary of PHoenix trial assessments published by Clinical Research & Innovation Office

Data Composition

The dataset contains several types of data for each patient:

- **Demographic:** (e.g. age) recorded at baseline
- **History:** (e.g. date of diagnoses) recorded at baseline
- **Intervention metadata:** (e.g. riociguat 5mg daily) Date-stamped records of medication regime, titrations and changes
- **Clinical assessments:**
 - **Functional test results** (e.g. WHO functional class), recorded at baseline and periodically during followup visits
 - **Patient-reported outcomes from questionnaires** (e.g. ED5D5L) collected at x intervals
- **Temporal device data:** (e.g blood pressure) physiological time-series from implantable devices

References

1. Pulmonary Hypertension: Intensification and Personalization of Combination Rx (PHoenix): A phase IV randomized trial for the evaluation of dose-response and clinical efficacy of riociguat and selexipag using implanted technologies
2. Remote monitored physiological response to therapeutic escalation and clinical worsening in patients with pulmonary arterial hypertension